PREDICTION CODE :

```
import numpy as np
import cv2
IMG_SIZE = 224
classes = [
'ABCA4',
'BEST1',
'EFEMP1',
'EYS',
'MYO7A',
'PRPH2',
'RPGR',
'RS1',
'TIMP3',
'USH2A'
]
image_path = "/content/retina_dataset/ABCA4/image44.png"
img = cv2.imread(image_path)
img = cv2.resize(img, (224,224))
img_array = img / 255.0
img_array = np.expand_dims(img_array, axis=0)
prediction = model.predict(img_array)
class_index = np.argmax(prediction)
predicted_class = classes[class_index]
confidence = prediction[0][class_index]
print("✅ Prediction:", predicted_class)
print("✅ Confidence:", confidence)
```

HEATMAP :

```
import tensorflow as tf
import matplotlib.pyplot as plt
```

```
last_conv_layer = model.get_layer("top_conv")
grad_model = tf.keras.models.Model(
    [model.inputs],
    [last_conv_layer.output, model.output]
)
with tf.GradientTape() as tape:
    conv_outputs, predictions = grad_model(img_array)
    class_channel = predictions[:, class_index]
grads = tape.gradient(class_channel, conv_outputs)
pooled_grads = tf.reduce_mean(grads, axis=(0,1,2))
conv_outputs = conv_outputs[0]
heatmap = conv_outputs @ pooled_grads[..., tf.newaxis]
heatmap = tf.squeeze(heatmap)
heatmap = tf.maximum(heatmap, 0) / tf.math.reduce_max(heatmap)
heatmap = heatmap.numpy()
plt.matshow(heatmap)
plt.show()
```

OVERLAY:

```
heatmap = cv2.resize(heatmap, (224,224))
heatmap = np.uint8(255 * heatmap)
heatmap = cv2.applyColorMap(
    heatmap,
    cv2.COLORMAP_JET
)
superimposed_img = heatmap * 0.4 + img
cv2.imwrite(
    "xai_output.jpg",
    superimposed_img
)
print("✅ XAI Image Saved")
```

PDF REPORT GENERATOR:

```
from reportlab.platypus import SimpleDocTemplate
from reportlab.platypus import Paragraph
from reportlab.platypus import Spacer
from reportlab.platypus import Image
from reportlab.lib.styles import getSampleStyleSheet
from datetime import datetime
styles = getSampleStyleSheet()
# Get current date
current_date = datetime.now().strftime("%d-%m-%Y %H:%M")
# Optional patient details
patient_name = "Sample Patient"
patient_id = "RP001"
model_name = "EfficientNetB0 Retina Model"
# Create PDF
pdf = SimpleDocTemplate("retina_detailed_report.pdf")
story = []
# Title
story.append(
    Paragraph(
        "Retinal Disease Detection Report",
        styles['Title']
    )
)
story.append(Spacer(1,20))
# Patient Details
story.append(
    Paragraph(
        f"<b>Patient Name:</b> {patient_name}",
        styles['Normal']
    )
)
```

```
story.append(
    Paragraph(
        f"<b>Patient ID:</b> {patient_id}",
        styles['Normal']
    )
)
story.append(
    Paragraph(
        f"<b>Date:</b> {current_date}",
        styles['Normal']
    )
)
story.append(Spacer(1,15))
# Prediction Results
story.append(
    Paragraph(
        "<b>Prediction Results</b>",
        styles['Heading2']
    )
)
story.append(
    Paragraph(
        f"<b>Predicted Disease:</b> {predicted_class}",
        styles['Normal']
    )
)
story.append(
    Paragraph(
        f"<b>Confidence Score:</b> {confidence:.2f}",
        styles['Normal']
    )
)
```

```
story.append(
    Paragraph(
        f"<b>Model Used:</b> {model_name}",
        styles['Normal']
    )
)
story.append(Spacer(1,15))
# Interpretation Section
story.append(
    Paragraph(
        "<b>Clinical Interpretation</b>",
        styles['Heading2']
    )
)

story.append(
    Paragraph(
        f"The AI system detected features consistent with "
        f"<b>{predicted_class}</b>. "
        f"The highlighted regions in the XAI image indicate "
        f"areas that contributed most to the prediction.",
        styles['Normal']
    )
)
story.append(Spacer(1,15))
# Recommendation Section
story.append(
    Paragraph(
        "<b>Recommendation</b>",
        styles['Heading2']
    )
)
```

```
story.append(
    Paragraph(
        "This automated result is intended for screening purposes only. "
        "Consult an ophthalmologist for professional diagnosis "
        "and clinical confirmation.",
        styles['Normal']
    )
)
story.append(Spacer(1,20))

# Original Image
story.append(
    Paragraph(
        "<b>Original Retina Image</b>",
        styles['Heading2']
    )
)
story.append(Spacer(1,10))
story.append(
    Image(
        image_path,
        width=250,
        height=250
    )
)
story.append(Spacer(1,20))
# XAI Image
story.append(
    Paragraph(
        "<b>XAI Visualization (Highlighted Regions)</b>",
        styles['Heading2']
    )
```

```
)
story.append(Spacer(1,10))
story.append(
    Image(
        "xai_output.jpg",
        width=250,
        height=250
    )
)
# Build PDF
pdf.build(story)
print("✅ Detailed PDF Report Generated")
```